

Natural Duality for Finitely Valued Algebras

Marco Abbadini

Université Catholique de Louvain (Belgium)

(Recording available at <https://youtu.be/1PCUjF2aBPg>)

AAL 2025 Conference

3 November 2025

Joint work with Adam Přenosil

Preprint: arXiv:2505.11490

The beauty of dualities

- ▶ Dualities offer a second viewpoint on algebraic structures:
geometric / spatial, rather than symbolic.
- ▶ Classical examples:
 - ▶ Stone: Boolean algebras $\leftrightarrow$ Stone spaces
 - ▶ Priestley: BDL $\leftrightarrow$ Priestley spaces
- ▶ An algebra corresponds to a structured space.

The power of dualities (beyond representation)

- ▶ Algebras in $\mathbf{ISP}(\mathbf{L})$ (the usual setting for such dualities) can already be **represented** as **L -valued functions** (with pointwise operations).
- ▶ A **duality** enriches this picture:
 - ▶ it **characterizes the canonical representations**,
 - ▶ it also represents **morphisms** in a natural way
$$f : \mathbf{A} \rightarrow \mathbf{B} \quad \longleftrightarrow \quad \text{continuous structure-preserving map,}$$
 - ▶ giving a **bijective correspondence** (both on objects and morphisms).
- ▶ This leads to a two-way **dictionary**:
 - ▶ products of algebras $\longleftrightarrow$ sums of spaces (“logarithmic” compression)
 - ▶ coproducts of algebras $\longleftrightarrow$ products of spaces (easier to describe)
 - ▶ congruences $\longleftrightarrow$ subspaces
 - ▶ free algebras $\longleftrightarrow$ powers of the dual of $\mathbf{Free}(1)$
 - ▶ algebraic questions $\longleftrightarrow$ **geometric/topological** ones

Our starting point: positive MV-algebras

- ▶ We were interested in obtaining a duality for **positive MV-algebras**:
 - ▶ the $\{\oplus, \odot, \vee, \wedge, 0, 1\}$ -subreducts of MV-algebras,
 - ▶ i.e. the quasivariety generated by $[0, 1]$ with these operations.
- ▶ Here, the natural dualizing algebra would be $[0, 1]$.
- ▶ Dualities of a similar flavour exist for MV-algebras:
 - ▶ Cignoli–Dubuc–Mundici (2003): locally finite MV-algebras.
Dualizing algebra: $[0, 1] \cap \mathbb{Q}$,
 - ▶ Cignoli–Marra (2012): weakly locally finite MV-algebras.
Dualizing algebra: $[0, 1]$.

- ▶ $[0, 1]$ is infinite.
- ▶ The general theory of natural dualities is well developed for **finite** dualizing algebras $\mathbf{L}$.
- ▶ There are extensions to some infinite $\mathbf{L}$, but they typically rely on equipping $\mathbf{L}$ with a **compact Hausdorff topology** and using it in the duality.
- ▶ The MV-dualities mentioned above follow a *different* pattern:
 - ▶ no topology is used on $\mathbf{L}$ (even for $\mathbf{L} = [0, 1]$),
 - ▶ but at a price: the duality applies not to all of $\mathbf{ISP}(\mathbf{L})$, but to a **restricted class** of algebras.
- ▶ **Goal:** provide a natural-duality framework that generalizes this MV-style phenomenon to other infinite untopologized dualizing algebras $\mathbf{L}$.

A clue from the MV side

- ▶ In both Cignoli–Dubuc–Mundici (2003) and Cignoli–Marra (2012), the duality works because elements of the algebras behave like **functions with finite range** in $\mathbf{L}$.
- ▶ Intuition: think of $a \in \mathbf{A}$ as an $\mathbf{L}$ -valued function on a Stone space, that takes only finitely many values in $\mathbf{L}$.
- ▶ This suggests that the right setting for infinite $\mathbf{L}$ is **finitely valued $\mathbf{L}$ -algebras**.

A restrictive but natural class: finitely valued algebras

Finitely valued $\mathbf{L}$ -algebra: an algebra $\mathbf{A}$ s.t. there is a set X and an embedding $\mathbf{A} \hookrightarrow \mathbf{L}^X$ s.t. each $f \in \mathbf{A}$ has **finite image in $\mathbf{L}$** .

I.e., it has *some* representation with functions of finite range. ($\mathbf{L}$ -algebra: an alg. in $\mathbf{ISP}(\mathbf{L})$.)

Under mild conditions (present in our duality result), this is equivalent to:

Canonically finitely valued $\mathbf{L}$ -algebra: an algebra $\mathbf{A} \in \mathbf{ISP}(\mathbf{L})$ s.t., for each $a \in \mathbf{A}$, the set $\{h(a) \mid h \in \text{hom}(\mathbf{A}, \mathbf{L})\}$ is finite.

I.e., the canonical representation $\mathbf{A} \hookrightarrow \mathbf{L}^{\text{hom}(\mathbf{A}, \mathbf{L})}$ is with functions of finite range.

Sanity checks:

- ▶ all finite powers $\mathbf{L}^n$ and their subalgebras are finitely $\mathbf{L}$ -valued;
- ▶ when $\mathbf{L}$ is finite, “finitely valued $\mathbf{L}$ -algebra” = “in $\mathbf{ISP}(\mathbf{L})$ ”.

Viewpoint via class operators

- ▶ For a dualizing algebra $\mathbf{L}$, standard natural dualities work with the whole $\mathbf{ISP}(\mathbf{L}) =$ algebras built from $\mathbf{L}$ by isomorphisms, subalgebras and arbitrary products.
- ▶ For an arbitrary (possibly infinite, non-topologized) $\mathbf{L}$, we replace $\mathbf{P}(\mathbf{L})$ (all powers $\mathbf{L}^X$) by the class of **finite-range powers**:

$$\mathbf{FinRng}(X, \mathbf{L}) := \{ f: X \rightarrow \mathbf{L} \mid f[X] \text{ is finite} \} \leq \mathbf{L}^X.$$

Define:

$$\mathbf{P}^{\text{fr}}(\mathbf{L}) := \{ \mathbf{FinRng}(X, \mathbf{L}) \mid X \text{ any set} \}.$$

- ▶ Our duality will apply to $\mathbf{ISP}^{\text{fr}}(\mathbf{L})$ (= the class of finitely valued $\mathbf{L}$ -algebras), not to $\mathbf{ISP}(\mathbf{L})$. (For $\mathbf{L}$ finite, they coincide.)

Assumptions on the dualizing algebra $\mathbf{L}$

Think of $\{0, 1\}$ (Bool. alg. / bdd. distr. lattice) or $[0, 1]$ (MV-algebra).

- ▶ $\mathbf{L}$ has at least **two distinct constant symbols**.

- ▶ $\mathbf{L}$ has only **trivial partial endomorphisms**.

i.e., for any $\mathbf{A} \leq \mathbf{L}$, the inclusion $\mathbf{A} \hookrightarrow \mathbf{L}$ is the unique homomorphism $\mathbf{A} \rightarrow \mathbf{L}$.

- ▶ $\mathbf{L}$ has a $(k+1)$ -ary **near-unanimity** term, with $k \geq 2$.

(E.g.: $\mathbf{L}$ has a lattice reduct; $\rightarrow$ majority term, i.e. ternary near-unanimity term).

Remarks.

- ▶ Under these hypotheses, “finitely valued algebras” = “canonically finitely valued algebras”. Moreover, the dual structure is purely relational + topological (no function symbols).

- ▶ The $(k+1)$ -ary **near-unanimity** term allows the dual space to be a Stone space + k -ary constraints: e.g., in Priestley spaces, the order is a set of binary constraints.

Duality theorem (finitely valued setting)

Main result (A., Přenosil)

Let $\mathbf{L}$ satisfy the assumptions on the previous slide. Then the category

$$\mathbf{ISP}^{\text{fr}}(\mathbf{L}) \quad (\text{finitely valued } \mathbf{L}\text{-algebras and homomorphisms})$$

*is dually equivalent to the category of k -ary **L-Priestley spaces** (and appropriate maps).*

In particular:

- ▶ if $k = 2$ (majority), the dual structure is driven by **binary** local constraints (Priestley-flavoured);
- ▶ if $\mathbf{L}$ is finite, $\mathbb{P}^{\text{fr}}(\mathbf{L}) = \mathbb{P}(\mathbf{L})$ and we recover the usual natural-duality scope.

k -ary $\mathbf{L}$ -Priestley spaces (definition)

A k -ary $\mathbf{L}$ -Priestley space consists of:

- ▶ a Stone space X ;
- ▶ for every $I \subseteq X$ with $|I| \leq k$, a **subalgebra** $\mathbf{A}_I \leq \mathbf{L}^I$, thought of as the set of *admissible $\mathbf{L}$ -valued local functions* (on I).

These data satisfy:

- ▶ **Separation:** for $x \neq y \in X$ there is $f \in \mathbf{A}_{\{x,y\}}$ s.t. $f(x) \neq f(y)$.
- ▶ **Extension:** for $I \subseteq X$ with $|I| \leq k$ and $f \in \mathbf{A}_I$, there is a continuous $g: X \rightarrow \mathbf{L}$ (with $\mathbf{L}$ discrete) s.t., for all $J \subseteq X$ with $|J| \leq k$, $g|_J \in \mathbf{A}_J$.

Comment. When $k = 2$ (majority), the structure is determined by **binary** constraints $\mathbf{A}_{\{x,y\}}$, echoing the Priestley paradigm.

Example: recovering Priestley when $L = 2$, $k = 2$

Let $\mathbf{L} = \mathbf{2} = \{0, 1\}$ (with $\wedge, \vee, 0, 1$), which has a majority term, so $k = 2$. For a Priestley space $(X, \leq)$ define, for each $I \subseteq X$ with $|I| \leq 2$,

$$\mathbf{A}_I := \{f : I \rightarrow \mathbf{2} \mid f \text{ is order-preserving}\} \leq \mathbf{2}^I.$$

Then $(X, (\mathbf{A}_I)_I)$ is a 2-ary $\mathbf{2}$ -Priestley space:

- **Separation:** if $x \neq y$, then either $x \not\leq y$ or $y \not\leq x$. In the former case, take $x \mapsto 1$ and $y \mapsto 0$, otherwise take $x \mapsto 0$ and $y \mapsto 1$.
- **Extension:** if $f \in \mathbf{A}_I$ with $|I| \leq 2$, then f extends to a continuous $g : X \rightarrow \mathbf{2}$ with $g|_J \in \mathbf{A}_J$ for all $|J| \leq 2$ (this is Priestley's separation axiom).

From k -ary $\mathbf{L}$ -Priestley spaces to finitely valued algebras

The functor maps a k -ary $\mathbf{L}$ -Priestley space $(X, (\mathbf{A}_I)_{I \in [X]^{\leq k}})$ to

$$\left\{ f: X \rightarrow \mathbf{L} \mid f \text{ cont. (with } \mathbf{L} \text{ discrete), } f|_I \in \mathbf{A}_I \text{ for all } I \in [X]^{\leq k} \right\}.$$

It is a finitely valued $\mathbf{L}$ -algebra: $X \text{ compact} + \mathbf{L} \text{ discrete} \Rightarrow$ a continuous $X \rightarrow \mathbf{L}$ has finite image.

E.g., in the Priestley case, we get

$$\{ f: X \rightarrow \mathbf{2} \mid f \text{ continuous and order-preserving} \},$$

i.e. the lattice of clopen upsets of $(X, \leq)$.

For the MV-algebras $\mathbf{L} = [0, 1] \cap \mathbb{Q}$ and $\mathbf{L} = [0, 1]$, our duality gives the dualities of Cignoli–Dubuc–Mundici for locally finite MV-algebras, and of Cignoli–Marra for weakly locally finite MV-algebras.

(This is after some further simplifications that make possible turning binary constraints into unary constraints in certain cases, which we describe in our preprint.)

Main result (A., Přenosil)

Let $\mathbf{L}$ be an algebra such that:

- ▶ $\mathbf{L}$ has at least **two distinct constant symbols**;
- ▶ $\mathbf{L}$ has only **trivial partial endomorphisms**.

i.e., for any $\mathbf{A} \leq \mathbf{L}$, the inclusion $\mathbf{A} \hookrightarrow \mathbf{L}$ is the unique homomorphism $\mathbf{A} \rightarrow \mathbf{L}$.

- ▶ $\mathbf{L}$ has a $(k+1)$ -ary **near-unanimity** term, with $k \geq 2$.

(E.g.: $\mathbf{L}$ has a lattice reduct; $\rightarrow$ majority term, i.e. ternary near-unanimity term).

Then the category

$\mathbf{ISP}^{\text{fr}}(\mathbf{L})$ (**finitely valued $\mathbf{L}$ -algebras** and homomorphisms)

is dual to the category of k -ary **$\mathbf{L}$ -Priestley spaces**.

Thank you!

ArXiv: *Abbadini, Přenosil,*
Duality for finitely valued algebras

